

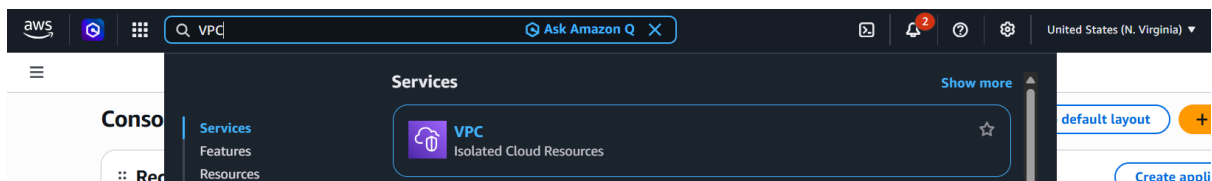
Creating a VPC and Subnet

To Begin with the Lab

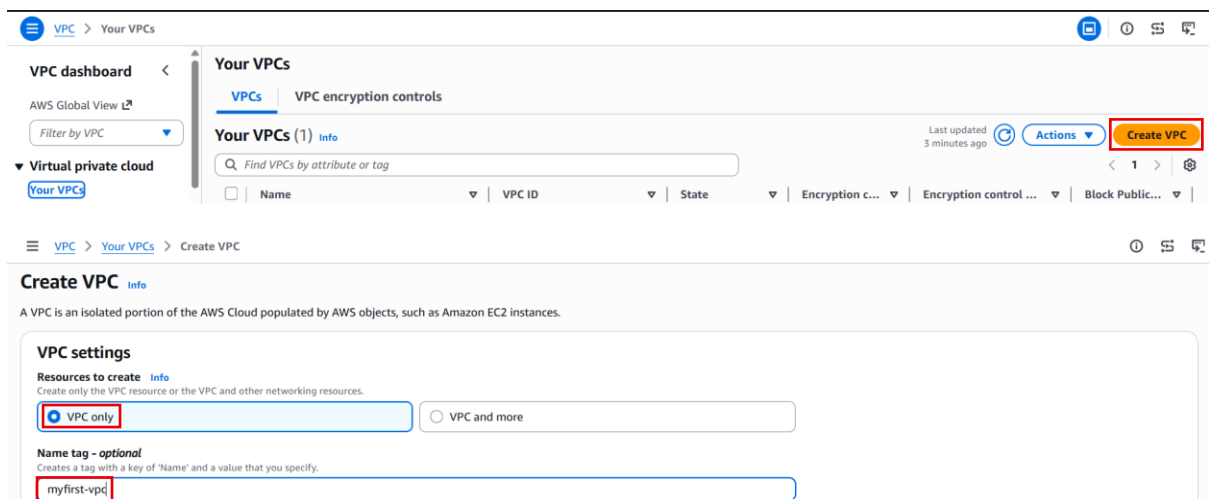
Summary of the Lab

This lab demonstrated how to create a custom Amazon VPC and a subnet using the AWS Management Console. A VPC was created with a specified IPv4 CIDR block, followed by the creation of a subnet within the VPC by selecting an Availability Zone and assigning a valid subnet CIDR block. The lab established the foundational network infrastructure for deploying AWS resources securely.

- Note: Check the region, all the resources belong to same region until said or prompted.
- Open the **AWS Management Console** and navigate to the **VPC** service.
- Go to VPC Dashboard.



- Click **Create VPC** and choose the **VPC only** option. Provide a meaningful name for the VPC, such as **myfirst-vpc**, so it can be easily identified while configuring other AWS services.



- Enter an IPv4 CIDR block for the VPC, such as **192.168.0.0/24**, to define the private IP address range that will be used by all resources created inside this VPC. After verifying the details, click **Create VPC**.

IPv4 CIDR block [Info](#)

☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

192.168.0.0/24

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)

☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy [Info](#)

Default

VPC encryption control (\$) [Info](#)

Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. [Additional charges apply](#)

☒ None
☐ Monitor mode
☐ Enforce mode

See which resources in your VPC are unencrypted but allow the creation of unencrypted resources.
Requires all resources, except exclusions, in your VPC to be encryption-capable and blocks creation of unencrypted resources.

- After the VPC is created successfully

You successfully created vpc-02a0440be172ac2c9 / myfirst-vpc

vpc-02a0440be172ac2c9 / myfirst-vpc [Actions](#)

Details [Info](#)

VPC ID vpc-02a0440be172ac2c9	State Available	Block Public Access Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-07bdad88c22547ed7	Main route table rtb-039041c66222da988
Main network ACL acl-0e1a390b090f74f45	Default VPC No	IPv4 CIDR 192.168.0.0/24	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 463646775279
Encryption control ID -	Encryption control mode -		

- open the **Subnets** section from the VPC dashboard and click **Create subnet**.

VPC > Subnets

VPC dashboard [AWS Global View](#)

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Subnets [Info](#)

Find subnets by attribute or tag

VPC = vpc-02a0440be172ac2c9 [Clear filters](#)

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
No matching resource found					

- Select the newly created VPC so that the subnet is associated with the correct network.

VPC > Subnets > Create subnet

Create subnet [Info](#)

VPC

VPC ID

Create subnets in this VPC.

vpc-02a0440be172ac2c9 (myfirst-vpc)

Q

vpc-099792461c7d0a616 (default)
172.31.0.0/16

vpc-02a0440be172ac2c9 (myfirst-vpc)
192.168.0.0/24

- Enter a name for the subnet, such as **Subnet-1**, and select an **Availability Zone** like **us-east-1a**. A subnet can exist in only one Availability Zone, so choose the required zone carefully based on your deployment requirements.

-

- Specify a CIDR block for the subnet that falls within the VPC's CIDR range, and then click **Create subnet**. This subnet will provide the network in which other resources can be launched.

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Subnet-1
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / use1-az4 (us-east-1a)

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
192.168.0.0/24

IPv4 subnet CIDR block
192.168.0.0/24 256 IPs

Tags - optional
[Remove](#)

- Subnet successfully created.

You have successfully created 1 subnet: subnet-04bad280d8c0e460e

Subnets (1) [Info](#)

Find subnets by attribute or tag

Subnet ID: subnet-04bad280d8c0e460e [Clear filters](#)

	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	Subnet-1	subnet-04bad280d8c0e460e	Available	vpc-02a0440be172ac2c9 myfi...	Off	192.168.0